# Session 4 – Assignment Answers

## 1. List all 6 phases of SDLC in order with one line about each.

1. Requirement Analysis: In this phase, the team collects and understands the user's needs and project requirements.
2. Design: The team plans the software architecture, database, user interface, and system structure.
3. Development: Developers write the code to build the software according to the design.
4. Testing: The software is tested to identify bugs and ensure that it works correctly.
5. Deployment: The completed software is released to users.
6. Maintenance: The team fixes bugs, improves performance, and updates the software after release.

## 2. Write 3 differences between Agile and Waterfall models.

| Agile Model | Waterfall Model |
|---|---|
| Development happens in small, iterative cycles. | Development follows a fixed sequence of phases. |
| Requirements can change during development. | Requirements are generally defined before development begins. |
| Testing and feedback happen continuously. | Testing usually takes place after the development phase. |

## 3. Name the 3 main Scrum roles and write one line about the responsibility of each.

1. Product Owner (PO): Defines product goals, manages the Product Backlog, and prioritizes work according to business and user needs.
2. Scrum Master: Helps the team follow Scrum practices and removes obstacles that affect progress.
3. Developers (Development Team): Build, test, and deliver a usable product increment during each Sprint.

## 4. What is a Sprint in Scrum? Mention its typical duration.

A Sprint is a fixed period during which the Scrum Team works to achieve a specific goal and produce a usable product increment.

Typical duration: 1–4 weeks, commonly 2 weeks.

## 5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

| Column | Meaning |
|---|---|
| To Do | Tasks that are ready to be started but have not yet been started. |
| In Progress | Tasks that are currently being worked on. |
| Done | Tasks that have been completed and meet the team's completion criteria. |
| Backlog | A list of pending features, tasks, and ideas that may be prioritized for future work. |